

SupportStream — Architecture Guide

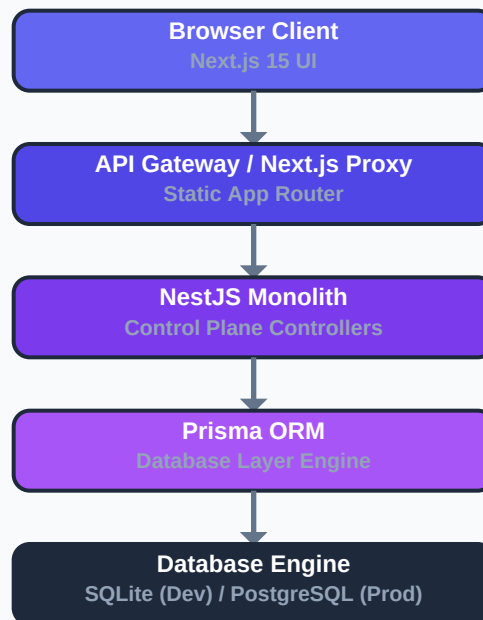
Enterprise Customer Support Ecosystem with Real-Time Video Assistance

1. System Vision & Problem Statement

Enterprise customer service centers require a secure, reliable, and high-performance video support architecture. Standard Peer-to-Peer (P2P) WebRTC models suffer from connection instability, IP address exposure, and a complete lack of recording/auditability. SupportStream addresses these gaps by decoupling the **Control Plane** (orchestrating session tokens, integrations, and workflows) from the **Data Plane** (mediasoup-based Selective Forwarding Unit SFU server processes). This enforces security, reliability, 5-minute client recovery intervals, and comprehensive operational visibility.

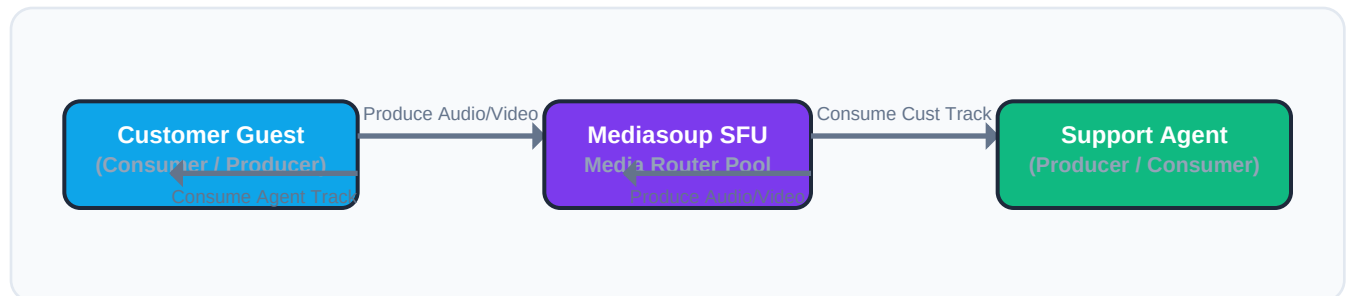
2. Full-Stack Control Flow & Data Access

This flow diagram illustrates the data flow of the SupportStream control plane, mapping how client-side interactions in the web browser propagate down to the backend API layer and write to the database engine.



3. SFU Media Server Architecture

SupportStream implements server-routed media routing using Mediasoup. Clients do not establish direct P2P connections; instead, they push audio, video, and screen sharing tracks as producers to the SFU worker process, which forwards them as consumers to the remote peer. This ensures maximum connection stability, low latency, and enables server-side recording.



4. Session Lifecycle State Machine

Support call sessions traverse an auditable state machine. This lifecycle allows SupportStream to enforce invite security rules, track customer wait times, verify link connections, and manage post-call processing like generating AI summaries and sync logs.



5. Outcome Sentiment & Ticket Escalation Flow

Post-session feedback is validated on the backend. When a customer submits a satisfaction score of 2 stars or fewer, an automated escalation workflow rule is triggered. The session outcome is set to ESCALATED, and the ticket reference is immediately routed to the department's Critical Escalations queue column for immediate manager attention.



6. Key Technology Stack Mapping

Component Layer	Technology	Architectural Responsibility
Frontend Client	Next.js 15 (React, Tailwind, Zustand)	Renders premium dark theme UI, device pre-check, screen-sharing hooks, and live dashboards.
Control Monolith	NestJS (TypeScript, WebSockets)	Manages session JWT states, ticket reference sequences, and triggers automation integrations.
SFU Media Server	Mediasoup (C++ Worker Sub-processes)	Handles high-performance WebRTC packet routing, stream encryption, and multi-track feeds.
Database ORM	Prisma Client (SQLite / PostgreSQL)	Provides type-safe access schemas, atomic transaction locks, and sequential ticket index generations.